Fault Type Classification from Simulated Power System Waveforms using 1D CNN

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https://github.com/hollyhan04/APS360_Project.git

Introduction

Transmission-line faults are among the most common disturbances in electric power systems. Accurate fault classification enables protection systems to isolate faulted equipment quickly, reducing equipment damage and improving overall system reliability. Traditional approaches often rely on manually designed signal-processing techniques and protection rules, whose performance may vary under different fault conditions.

Recent advances in deep learning have made it possible to learn fault-related patterns directly from voltage and current measurements. In particular, convolutional neural networks (CNNs) have demonstrated strong performance on a variety of time-series classification problems by automatically extracting informative features from raw signals. The goal of this project is to develop a 1D CNN that classifies transmission-line faults using short post-fault windows of three-phase voltage and current waveforms, distinguishing among multiple fault types and normal operating conditions. Its performance will be compared with a vanilla RNN baseline [1].

Illustration

Figure 1 summarizes the proposed workflow. Fault scenarios are first simulated in the IEEE/WSCC 9-bus system to generate labeled voltage and current waveforms [2].. These waveforms are then partitioned into training, validation, and test sets and used to train both the proposed 1D CNN and a vanilla RNN baseline. Model performance is compared using accuracy, macro-F1 score, and confusion matrices.

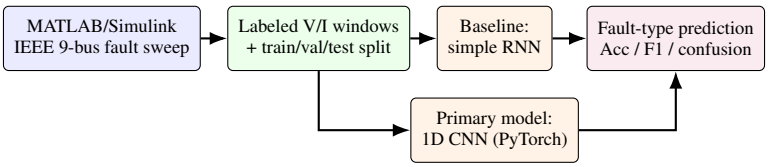


Figure 1: Project pipeline

Background & Related Work

Fault classification has been an active area of research in power-system protection because the ability to identify fault types quickly and accurately is essential for reliable grid operation. Early studies explored artificial neural networks (ANNs) as alternatives to conventional rule-based protection schemes. These works demonstrated that neural networks could successfully classify transmission-line faults directly from voltage and current measurements, establishing machine learning as a promising approach for fault diagnosis [3, 4].

As deep learning techniques became more widely adopted, researchers began moving away from manually engineered signal features toward models capable of learning representations directly from waveform data. Chen et al. proposed a convolutional sparse autoencoder that learned fault-related features automatically from electrical measurements [5]. Building on this trend, Fahim et al.

developed a self-attention CNN architecture for transmission-line fault classification, demonstrating the effectiveness of convolutional feature extraction for power-system disturbances [6].

Similar approaches have also been applied to related power-system applications. Guo et al. combined wavelet transforms with CNNs for earth-fault detection, while recent surveys have highlighted deep learning as an increasingly important tool for power-system monitoring, protection, and fault diagnosis [7, 8]. Collectively, these studies indicate a clear evolution from traditional ANN-based classifiers toward deep learning models that automatically learn informative features from raw electrical waveforms. Motivated by these findings, this project investigates the effectiveness of a 1D CNN for transmission-line fault classification using simulated voltage and current measurements from the IEEE/WSCC 9-bus system.

Data Processing

Training data will be generated in MATLAB/Simulink using the IEEE/WSCC 9-bus system [2]. Multiple fault scenarios will be simulated, including single-line-to-ground, line-to-line, double-line-to-ground, three-phase faults, and normal operating conditions. Simulations will vary fault location, fault resistance, and inception angle to improve dataset diversity.

For each event, three-phase voltages and line currents will be recorded and extracted into fixed-length windows following fault inception. The resulting multichannel waveforms will be normalized and resized to a common length through truncation or zero-padding. The dataset will then be divided into training, validation, and test sets while ensuring that closely related simulation scenarios do not appear in multiple splits.

The target dataset size is approximately 4,000–6,000 labeled samples, which is sufficient for training and evaluating deep-learning models within the scope of this project.

Architecture

The primary model will be a 1D convolutional neural network implemented in PyTorch. The network will take multichannel voltage and current waveforms as input and use several convolutional layers to extract fault-related temporal features. The learned representations will then be passed through fully connected layers to produce fault-type predictions. Hyperparameters such as network depth, kernel size, learning rate, and dropout rate will be selected using validation-set performance. Training will use cross-entropy loss and the Adam optimizer [9, 10]. If time permits, an ablation study will be conducted to investigate the contribution of key design choices, such as the inclusion of current measurements or the effect of different input-channel combinations [11].

Baseline Model

A vanilla RNN implemented in PyTorch will serve as the baseline model [12]. The RNN will be trained using the same dataset partitions, loss function, and evaluation metrics as the CNN to ensure a fair comparison. The proposed CNN will be considered successful if it achieves higher test accuracy and macro-averaged F1 score than the RNN baseline [1]. Confusion matrices will also be used to identify fault categories that are frequently misclassified.

Ethical Considerations

The dataset is fully simulated, so no personal or customer data are collected. However, training exclusively on simulated data introduces a risk that the model may not generalize to real power systems with different network topologies, protection schemes, noise characteristics, and operating conditions. In practice, incorrect fault classification could contribute to delayed fault isolation or unnecessary outages. Therefore, this work will be treated as a research prototype rather than a deployable protection system, and its limitations will be clearly documented. In addition, any discussion of model weaknesses or detection boundaries will be presented only in the context of improving system reliability and not as guidance for compromising power-system infrastructure.

References

[1] Marina Sokolova and Guy Lapalme. A systematic analysis of performance measures for classification tasks. *Information Processing & Management*, 45(4):427–437, 2009.

[2] Illinois Center for a Smarter Electric Grid (ICSEG). Wscc 9-bus system. `https://icseg.iti.illinois.edu/wscc-9-bus-system/`, 2013. Classic 9-bus test system used for fault studies.

[3] Thomas Dalstein and Bernd Kulicke. Neural network approach to fault classification for high speed protective relaying. *IEEE Transactions on Power Delivery*, 10(2):1002–1011, 1995.

[4] Majid Jamil, Sanjeev Kumar Sharma, and Rajveer Singh. Fault detection and classification in electrical power transmission system using artificial neural network. *SpringerPlus*, 4(1):334, 2015.

[5] Kunjin Chen, Jun Hu, and Jinliang He. Detection and classification of transmission line faults based on unsupervised feature learning and convolutional sparse autoencoder. *IEEE Transactions on Smart Grid*, 9(3):1748–1758, 2018.

[6] Shahriar Rahman Fahim, Yeadul Sarker, Subrata K. Sarker, Md. Rafiqul Islam Sheikh, and Sajal K. Das. Self attention convolutional neural network with time series imaging based feature extraction for transmission line fault detection and classification. *Electric Power Systems Research*, 187:106437, 2020.

[7] Mou-Fa Guo, Xiao-Dong Zeng, Duan-Yu Chen, and Nien-Che Yang. Deep-learning-based earth fault detection using continuous wavelet transform and convolutional neural network in resonant grounding distribution systems. *IEEE Sensors Journal*, 18(3):1291–1300, 2018.

[8] A. K. Ozcanli, F. Yaprakdal, and M. Baysal. Deep learning methods and applications for electrical power systems: A survey of the state-of-the-art. *Journal of Modern Power Systems and Clean Energy*, 8(4):627–643, 2020.

[9] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)*, 2015.

[10] Nitish Srivastava, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. Dropout: A simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15(56):1929–1958, 2014.

[11] Richard Meyes, Melanie Lu, Constantin Waubert de Puiseau, and Tobias Meisen. Ablation studies in artificial neural networks. *arXiv preprint arXiv:1901.08644*, 2019.

[12] Jeffrey L. Elman. Finding structure in time. *Cognitive Science*, 14(2):179–211, 1990.